

RSVP: Resource Reservation Protocol

Providing QoS by Reserving Resources in the Network

Objective

The objective of this lab is to study the Resource Reservation Protocol (RSVP) as a part of the Integrated Services approach to providing Quality of Service (QoS) to individual applications or flows.

Overview

For many years, packet-switched networks have offered the promise of supporting multimedia applications, that is, those that combine audio, video, and data. Audio and video applications are examples of *real-time* applications. The best-effort model, in which the network tries to deliver your data but makes no promises and leaves the “cleanup operation” to the edges, is not sufficient for real-time applications. What we need is a new service model—one in which applications that need better assurances can request such service from the network. The network may then respond by providing an assurance that it will do better, or perhaps by saying that it cannot promise anything better at the moment. A network that can provide different levels of service is often said to support QoS.

Two approaches have been developed to provide a range of QoS: Integrated Services and Differentiated Services. The Resource Reservation Protocol follows the Integrated Services approach, whereby QoS is provided to individual applications or flows. The Differentiated Services approach provides QoS to large classes of data or aggregated traffic.

While connection-oriented networks have always needed some sort of setup protocol to establish the necessary virtual circuit state in the routers, connectionless networks like the Internet have had no such protocols. One of the key assumptions underlying RSVP is that it should not detract from the robustness that we find in the Internet. Therefore, RSVP uses the idea of *soft state* in the routers. Soft state—in contrast to the hard state found in connection-oriented networks—does not need to be explicitly deleted when it is no longer needed. Instead, it times out after some fairly short period if it is not periodically refreshed. RSVP adopts the *receiver-oriented* approach—the receivers keep track of their own resource requirements, and they periodically send refresh messages to keep the soft state in place.

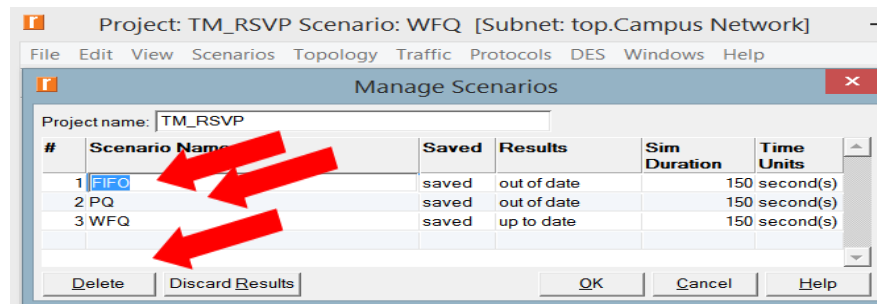
In this lab you will set up a network that carries real-time applications and that utilizes RSVP to provide QoS to one of these applications. You will study how RSVP contributes to the performance of the application that makes use of it.

Procedure

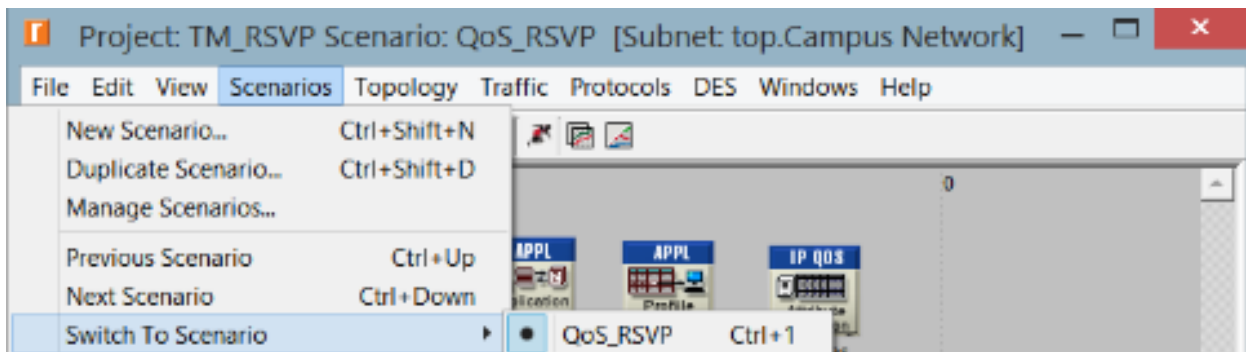
Create a New Project

1. Start **Riverbed Modeler Academic Edition** ⇒ Choose **Open** from the **File** menu.
2. Select the project you created in Lab 9: **<your initials>_Queues** ⇒ Click **OK**.
3. From the **File** menu, choose **Save As** ⇒ Rename the project to **<your initials>_RSVP** ⇒ Click **OK**.
4. From the **Scenarios** menu, choose **Manage Scenarios** ⇒ Click on **FIFO** ⇒ Click **Delete** ⇒ Click on **PQ** ⇒ Click **Delete**.

The idea of the **FQ** (fair queuing) discipline is to maintain a separate queue for each flow currently being handled by the router. The router then services these queues in a round-robin manner. **WFQ** allows a weight to be assigned to each flow (queue). This weight effectively controls the percentage of the link's bandwidth each flow will get. We could use the **ToS** (Type of Service) field in the IP header to identify that weight.



5. Click on **WFQ** and rename it to **QoS_RSVP** ⇒ Click **OK**.
6. Make sure that you have only one scenario in your project named **QoS_RSVP**. The following figure shows one way to check for the available scenarios in the project.



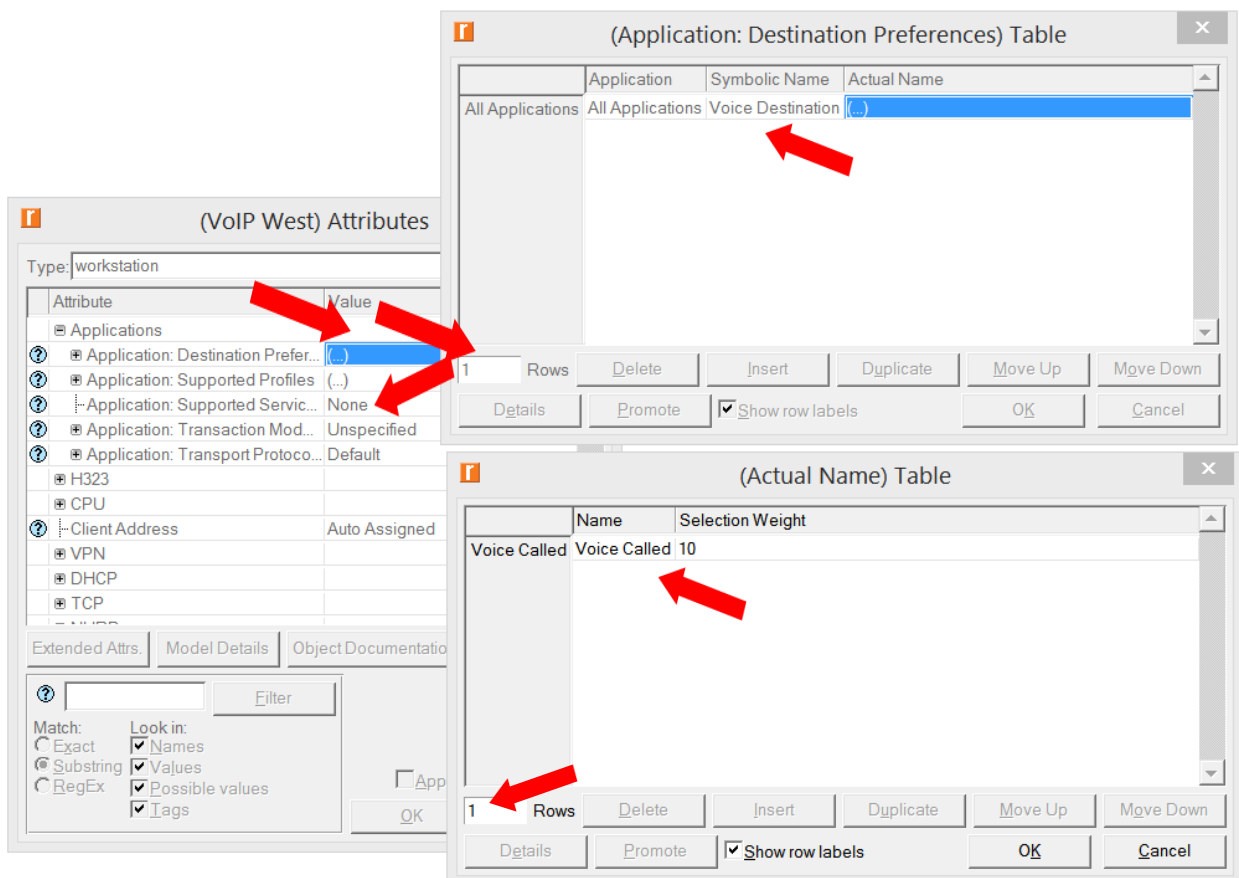
7. Save your project.

Configure the Network

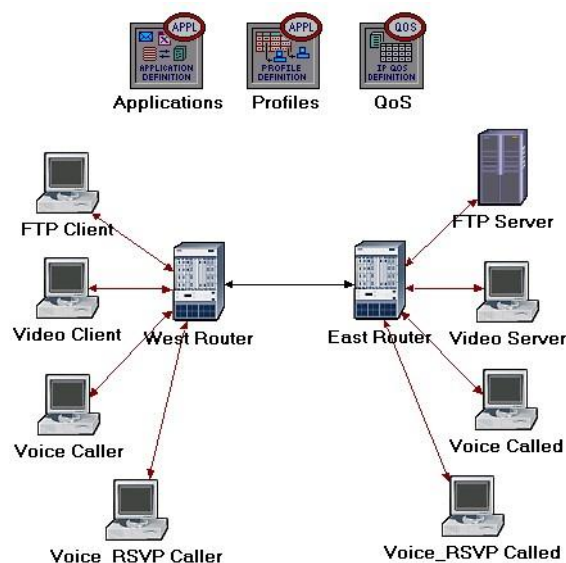
Add More VoIP Nodes:

In this project we will set up the two VoIP nodes so that one will always be the *Caller* party and the other will be the *Called* party. In addition, we will add two new VoIP *Caller* and *Called* nodes. These new nodes will utilize RSVP to reserve their required resources through the network.

1. Right-click on the **VoIP East** node ⇒ **Edit Attributes** ⇒ Rename the node to **Voice Called** ⇒ Assign **None** to the **Application: Supported Profiles** attribute inside **Applications** ⇒ Assign **Voice Called** to the **Client Address** attribute ⇒ Click **OK**.
2. Right-click on the **VoIP West** node ⇒ **Edit Attributes**.
 - i. Rename the node to **Voice Caller**.
 - ii. Assign **None** to the **Application: Supported Services** attribute inside **Applications**.
 - iii. Edit the value of the **Application: Destination Preferences** attribute inside **Applications** ⇒ Set **Rows** to 1 ⇒ Assign **Voice Destination** to the **Symbolic Name** of the new row ⇒ Edit the **Actual Name** attribute ⇒ Set **Rows** to 1 ⇒ Assign **Voice Called** to the **Name** attribute of the new row as shown.
 - iv. Click **OK** three times.



3. Click on the **Voice Called** node to select it ⇒ From the **Edit** menu, select **Copy** ⇒ From the **Edit** menu, select **Paste** (alternatively, use the standard keyboard shortcuts, **Ctrl-C** and **Ctrl-V**).
 - i. Locate the new node somewhere below the **Voice Called** node on the screen ⇒ Connect the new node to the **East Router** using a **10BaseT** link.
 - ii. Right-click on the new node ⇒ **Edit Attributes**.
 - iii. Click on the **ethernet_wkstn** value of the **model** attribute ⇒ Select **Edit** ⇒ Select the **ethernet_wkstn_adv** model.
 - iv. Rename it to **Voice_RSVP Called** ⇒ Assign **Voice_RSVP Called** to its **Client Address** attribute.
 - v. Click **OK**.
4. Copy and paste the **Voice Caller** node.
 - i. Locate the new node somewhere below the **Voice Caller** node ⇒ Connect the new node to the **West Router** using a **10BaseT** link.
 - ii. Right-click on the new node ⇒ **Edit Attributes**.
 - iii. Click on the **ethernet_wkstn** value of the **model** attribute ⇒ Select **Edit** ⇒ Select the **ethernet_wkstn_adv** model.
 - iv. Rename it to **Voice_RSVP Caller**.
 - v. Edit the **Application: Destination Preferences** attribute inside **Applications** ⇒ Open the **Actual Name** table by clicking in the value field of **Actual Name** ⇒ Assign **Voice_RSVP Called** to the **Name** attribute. (If you don't see **Voice_RSVP Called** in the list, please repeat step 3.)
 - vi. Click **OK** three times.
5. Rename the **Queues** node in the project to **QoS**. Your project should look like the following diagram.
6. Save your project.



Define the Data Flow:

Here, we will define the data flow characteristics of the voice traffic in the network. The sender's RSVP module periodically sends RSVP Path messages that uses the data flow characteristics to describe the traffic generated by the sender. When the receiver's RSVP module receives the Path message, the receiver host application checks the characteristics of the requested data flow and decides if resources should be reserved. Once a decision is made to request network resource reservation, the host application sends a request to the local RSVP module to assist in the reservation setup. The receiver's RSVP module then carries the request as Resv messages to all nodes along the reverse data path to the sender.

The flow is defined by its required bandwidth and buffer size. Bandwidth is set to be the *token bucket rate* in the flow specification of the Path and Resv messages. The buffer size represents the amount of the application "bursty" data to be buffered. It specifies the *token bucket size* that will be set in the Path or Resv messages for the session.

1. Right-click on the **QoS** node ⇒ **Edit Attributes**.
 - i. Expand the **RSVP Flow Specification** hierarchy and its **default** hierarchy ⇒ Set **Name** to **RSVP_Flow** ⇒ Assign **50,000** to the **Bandwidth (bytes/sec)** attribute ⇒ Assign **10,000** to the **Buffer Size (bytes)** attribute.
 - ii. Expand the **RSVP Profiles** hierarchy and its **default** hierarchy ⇒ Set **Profile Name** to **RSVP_Profile**.
 - iii. Click **OK** and then save your project.

The screenshot shows the "(QoS) Attributes" dialog box. The "Type" is set to "Utilities". The "Attribute" table is as follows:

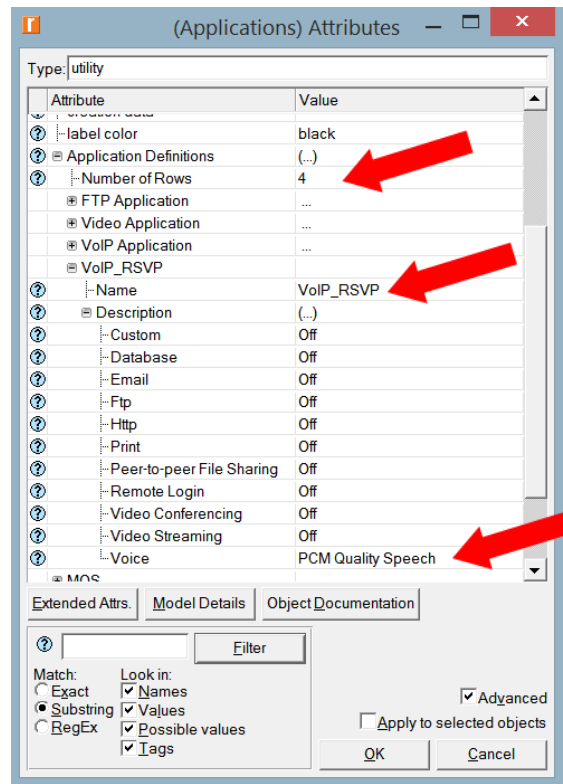
Attribute	Value
Priority Queuing Profiles	Standard Schemes
RSVP Flow Specification	(...)
Number of Rows	1
RSVP_Flow	
Name	RSVP_Flow
Bandwidth (bytes/sec)	50,000
Buffer Size (bytes)	10,000
RSVP Profiles	(...)
Number of Rows	1
Default	
Profile Name	RSVP_Profile
Threshold (bytes/sec)	None

At the bottom, there are tabs for "Extended Attrs.", "Model Details", and "Object Documentation". Below these are search filters: "Match" (Exact, Substring, RegEx) and "Look in:" (Names, Values, Possible values, Tags). There is also a checkbox for "Advanced" and "Apply to selected objects".

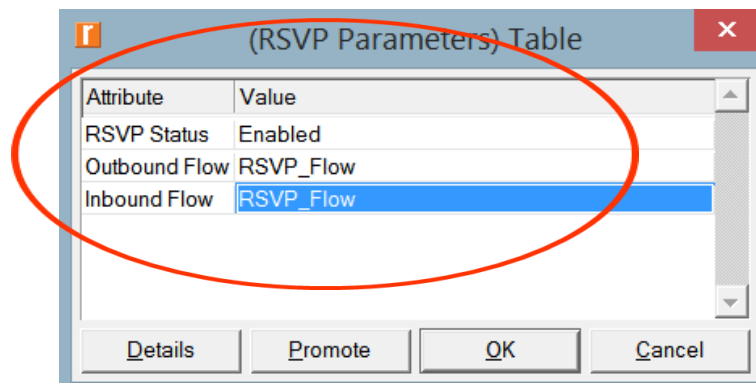
Configure the Application:

Here we will create a VoIP application that utilizes the RSVP flow specifications we configured.

1. Right-click on the **Applications** node ⇒ **Edit Attributes** ⇒ Expand the **Applications Definitions** hierarchy ⇒ Set **rows** to 4 (to add a fourth row to the **Application Definitions** attribute).
 - i. Name and set the attributes of row 4 as shown:



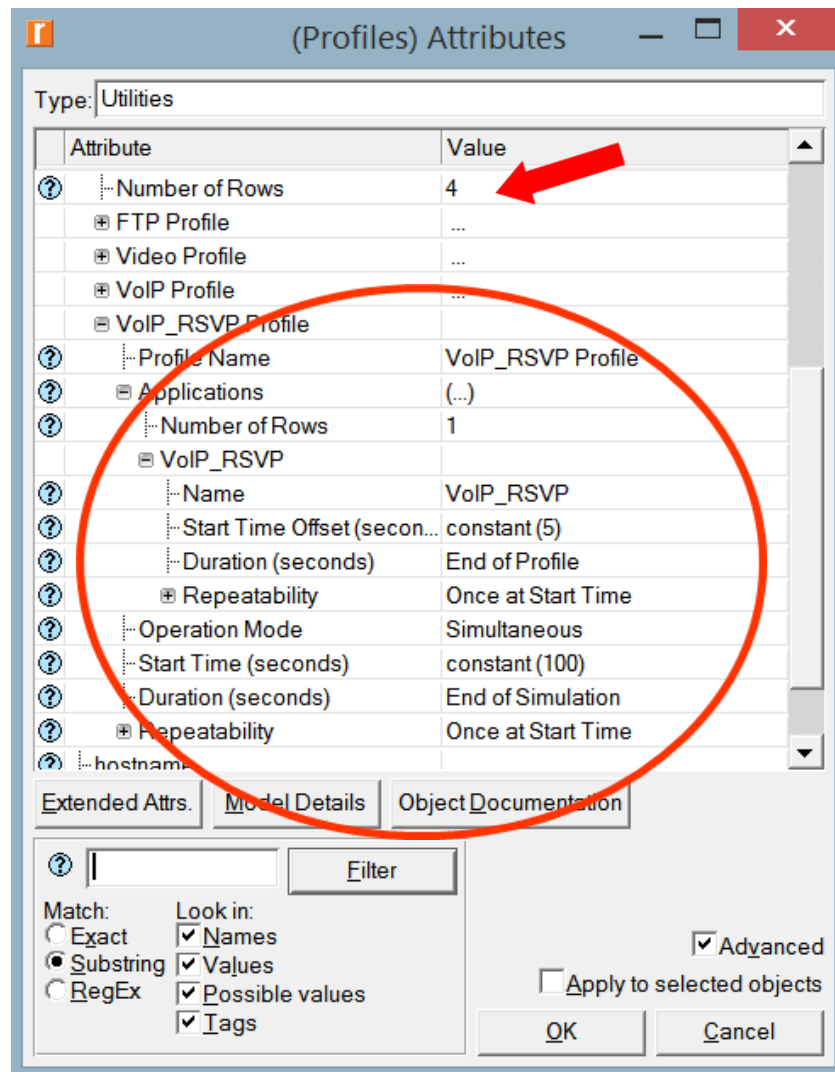
- ii. Click on the **PCM Quality Speech** value (shown above) ⇒ Select **Edit** ⇒ Edit the value of the **RSVP Parameters** attribute ⇒ Assign the following values (recall that we defined the **RSVP_Flow** in the QoS node, so if you don't see it on the list, repeat Data Flow step) ⇒ Click **OK** three times.



Note that the characteristics of the **Outbound Flow** are carried in the Path messages to be sent from sender to receiver, and the characteristics of the **Inbound Flow** parameters are carried in the Resv messages to be sent from the receiver to the sender.

Configure the Profile:

1. Right-click on the **Profiles** node ⇒ **Edit Attributes** ⇒ Expand the **Profile Configuration** hierarchy ⇒ Set **rows** to 4 (to add a fourth row to the **Profile Configuration** attribute) ⇒ Name and set the attributes of row 4 as shown:

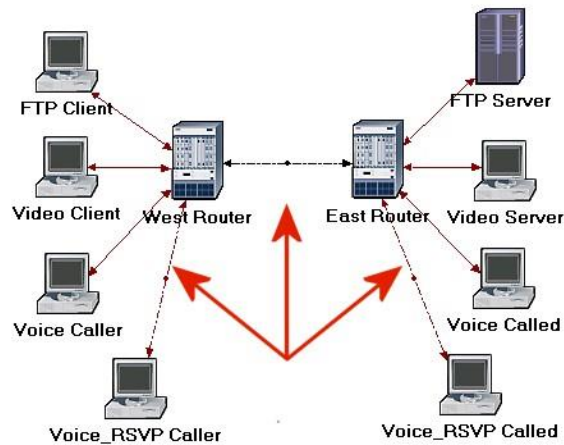


2. Click **OK** and then save your project.

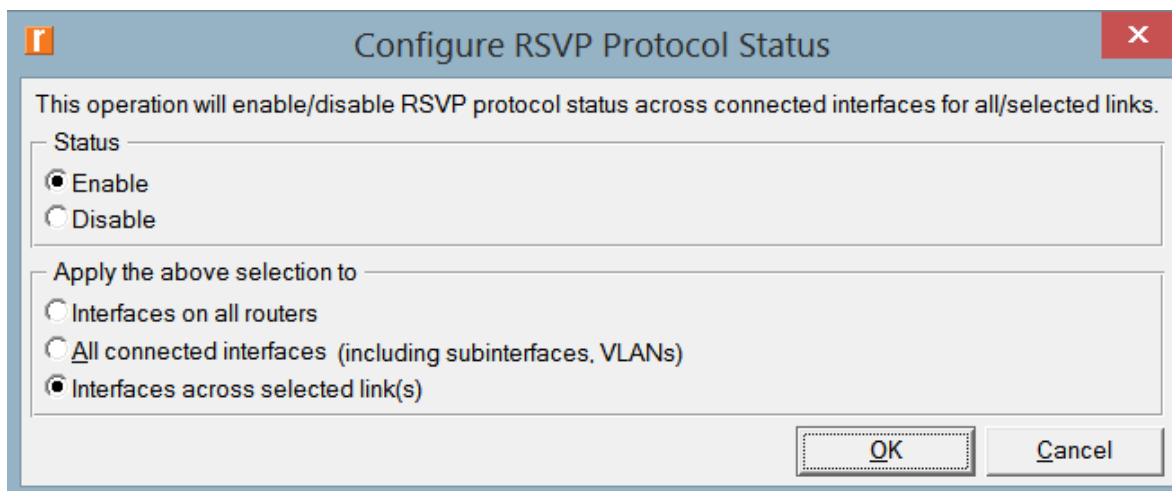
Configure the Interfaces:

Riverbed Modeler supports RSVP on a per-interface basis; RSVP can be enabled or disabled for each node's interface.

1. Simultaneously select (shift + left-click) the following three links:



2. From the **Protocols** menu, select **RSVP** ⇒ Select **Configure Interface Status...** ⇒ Make the selections shown below in the configuration dialog box ⇒ Click **OK** and then save your project.



The above process enables RSVP on all interfaces along the path between the two Voice parties that need to utilize RSVP.

Maximum Bandwidth Per Flow specifies the amount of reservable bandwidth that can be allocated to a single flow.

3. From the **Protocols** menu, select **RSVP** ⇒ Select **Configure Interface based per Flow Reservable Bandwidth...** ⇒ Make the selections shown below in the configuration dialog box ⇒ Click **OK** and then save your project.

Configure Per-Flow Reservable Bandwidth

This operation will configure the maximum bandwidth that a flow can reserve on an interface as a percentage of the connecting link or in terms of bits/sec.

Flow reservable BW: 75

Configure specified bandwidth as

☒ Percentage of link bandwidth
☐ Bits/second

Apply the above selection to

☐ All connected interfaces (including subinterfaces, VLANs)
☒ Interfaces across selected link(s)

OK Cancel

Configure the Hosts and Routers:

In Riverbed Modeler, the RSVP process runs only in IP-enabled nodes. The advanced versions (*_adv) of those node models must be used, as we did already, to configure RSVP-related parameters. In addition, the RSVP model in Riverbed Modeler requires either WFQ or custom queuing schemes.

1. Right-click on the **Voice_RSVP Caller** node ⇒ **Edit Attributes**.
 - i. Expand the **Application: Supported Profiles** hierarchy inside **Applications** and its **row 1** hierarchy (should be **VoIP Profile** if you have not changed it from the last one) ⇒ Assign **VoIP_RSVP Profile** to the **Profile Name** attribute (if you don't see it on the list, please repeat Profile step).
 - ii. Expand the **Application: RSVP Parameters** hierarchy inside **Applications** ⇒ Expand its **Voice** hierarchy ⇒ **Enable** the **RSVP Status** ⇒ Expand the **Profile List** hierarchy ⇒ Assign to the **Profile** attribute of **Default** the value **RSVP_Profile**.

Type: workstation

Attribute	Value
[-] Voice	(...)
[-] RSVP Status	Enabled
[-] Profile List	(...)
[-] Number of Rows	1
[-] RSVP_Profile	
[-] Profile	RSVP_Profile

- iii. Expand the **IP QoS Parameters** hierarchy inside **IP** ⇒ Expand its **Interface Information** hierarchy ⇒ Assign **1** to the **Number of Rows** value ⇒ Set Name to **IF0** ⇒ Expand the **QoS Scheme** hierarchy ⇒ Assign **1** to the **Number of Rows** value ⇒ Assign **WFQ (Class Based)** to the **Type** attribute ⇒ Assign **ToS Based** to the **Name** attribute.

Type of Service (ToS) is assigned to the IP packets. It represents a session attribute that allows packets to be provided the appropriate service in the IP queues.

Attribute	Value
[-] IP	
[-] Adaptive Service Configuration	Not Configured
[-] IP Host Parameters	(...)
[-] IP Processing Information	(...)
[-] IP QoS Parameters	(...)
[-] Interface Information	(...)
[-] Number of Rows	1
[-] IF0	
[-] Name	IF0
[-] QoS Scheme	(...)
[-] Number of Rows	1
[-] WFQ (Class Based)	
[-] Type	WFQ (Class Based)
[-] Name	ToS Based

Maximum Reservable BW specifies the percentage of the bandwidth of the connected link that RSVP can reserve on the interface.

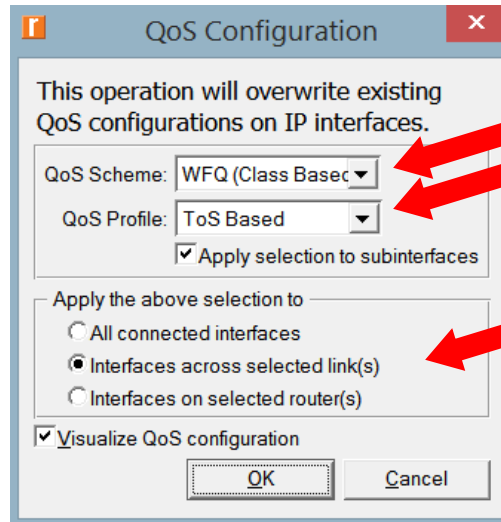
- iv. Expand the **RSVP Protocol Parameters** hierarchy inside **RSVP** ⇒ Expand the **Interface Information** hierarchy. (You should notice that the word *Enabled* is listed in the summary line. When you expand it, you will see that it is the value of **RSVP Status**. If *Enabled* is not listed, go back to the *Configure the Interfaces* steps.) ⇒ Expand the hierarchy of the row of that interface ⇒ Assign **75%** to the **Maximum Reservable BW** attribute as shown:

Attribute	Value
Interface Information (1 Row) (...)	
IF0	
Name	IF0
RSVP Status	Enabled
Maximum Reservable ...	75%

- v. Click **OK** and then save your project.
2. Right-click on the **Voice_RSVP Called** node ⇒ **Edit Attributes**.
 - i. Edit the **Application: Supported Services** attribute inside **Applications**. The *Application: Supported Services Table* will popup ⇒ In that table, replace the **VoIP Application** with **VoIP_RSVP** and click **OK**.
 - ii. Expand the **Application: RSVP Parameters** hierarchy inside **Applications** ⇒ Expand its **Voice** hierarchy ⇒ **Enable** the **RSVP Status** ⇒ Expand the **Profile List** hierarchy ⇒ Edit the value of the **Profile** attribute of **default** and write down **RSVP_Profile**.
 - iii. Expand the **IP QoS Parameters** hierarchy inside **IP** ⇒ Expand its **Interface Information** hierarchy ⇒ Assign **1** to the **Number of Rows** value ⇒ Set Name to **IF0** ⇒ Expand the **QoS Scheme** hierarchy ⇒ Assign **1** to the **Number of Rows** value ⇒ Assign **WFQ (Class Based)** to the **Type** attribute ⇒ Assign **ToS Based** to the **Name** attribute.
 - iv. Expand the **RSVP Protocol Parameters** hierarchy inside **RSVP** ⇒ Expand the **Interface Information** hierarchy. (You should notice that the word *Enabled* is listed in the summary line. When you expand it, you will see that it is the value of **RSVP Status**. If *Enabled* is not listed, go back to the *Configure the Interfaces* steps.) ⇒ Expand the hierarchy of the row of that interface ⇒ Assign **75%** to the **Maximum Reservable BW** attribute as shown above.
 - v. Click **OK** and then save your project.

3. Right-click on the **East Router** node ⇒ **Edit Attributes**.
 - i. Click on the **Ethernet4_slip8_gtwy** value of the **model** attribute ⇒ Select **Edit** ⇒ Select the **Ethernet4_slip8_gtwy_adv** model.
 - ii. Expand the **RSVP Protocol Parameters** hierarchy inside **RSVP** ⇒ Expand the **Interface Information** hierarchy. (You should notice that the **RSVP Status** of two interfaces, which are connected to the **West Router** and the **Voice_RSVP Called** node, are *Enabled*. If not, go back to the *Configure the Interfaces* steps.) ⇒ Expand the hierarchies of the rows of these two interfaces ⇒ Assign **75%** to the **Maximum Reservable BW**.
 - iii. Expand the **IP QoS Parameters** hierarchy inside **IP** ⇒ Expand its **Interface Information** hierarchy ⇒ Assign **2** to the **Number of Rows** value ⇒ Set Name to of the rows as the same two interfaces you configured in the previous step (step ii) ⇒ Expand the **QoS Scheme** hierarchy ⇒ Assign **1** to the **Number of Rows** value ⇒ Assign **WFQ (Class Based)** to the **Type** attribute ⇒ Assign **ToS Based** to the **Name** attribute.
 - iv. Click **OK** and then save your project.
4. Right-click on the **West Router** node ⇒ **Edit Attributes**.
 - i. Click on the **Ethernet4_slip8_gtwy** value of the **model** attribute ⇒ Select **Edit** ⇒ Select the **Ethernet4_slip8_gtwy_adv** model.
 - ii. Expand the **RSVP Protocol Parameters** hierarchy inside **RSVP** ⇒ Expand the **Interface Information** hierarchy. (You should notice that the **RSVP Status** of two interfaces, which are connected to the **West Router** and the **Voice_RSVP Caller** node, are *Enabled*. If not, go back to the *Configure the Interfaces* steps.) ⇒ Expand the hierarchies of the rows of these two interfaces ⇒ Assign **75%** to the **Maximum Reservable BW**.
 - iii. Expand the **IP QoS Parameters** hierarchy inside **IP** ⇒ Expand its **Interface Information** hierarchy ⇒ Assign **2** to the **Number of Rows** value ⇒ Set Name to of the rows as the same two interfaces you configured in the previous step (step ii) ⇒ Expand the **QoS Scheme** hierarchy ⇒ Assign **1** to the **Number of Rows** value ⇒ Assign **WFQ (Class Based)** to the **Type** attribute ⇒ Assign **ToS Based** to the **Name** attribute.
 - iv. Click **OK** and then save your project.

5. Click on the link connecting the **East** and **West** routers to select it ⇒ From the **Protocols** menu choose **IP** → **QoS** → **Configure QoS**. Make sure the selected items are as shown in the following *QoS Configuration* dialog box ⇒ Click **OK**.



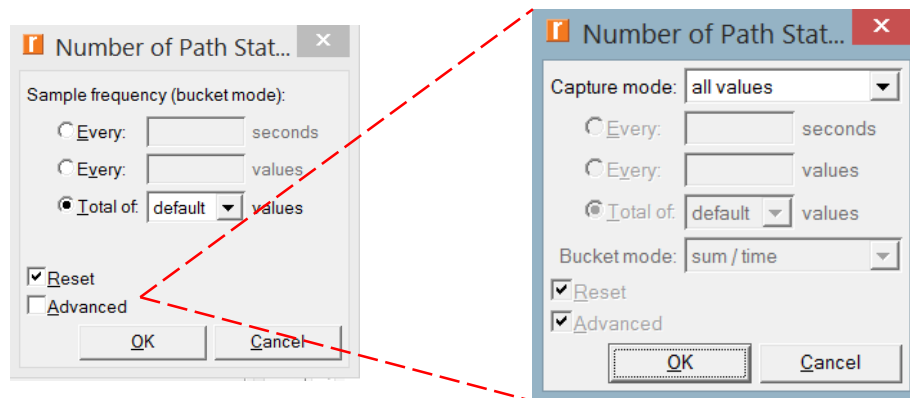
Note: Since the **Visualize QoS Configuration** radio button is checked, the link is colored based on the QoS scheme used.

Choose the Statistics

We will select statistics from three different nodes:

Voice_RSVP Caller Statistics:

1. Right-click on the **Voice_RSVP Caller** node and select **Choose Individual Statistics** from the pop-up menu.
2. Expand the **RSVP** hierarchy and select **Number of Path States**.
3. Right-click on the **Number of Path States** statistic ⇒ Select **Change Draw Style** from the pop-up menu ⇒ Choose **bar chart**.
4. Right-click on the **Number of Path States** statistic ⇒ Select **Change Collection Mode** from the pop-up menu ⇒ Check the **Advanced** checkbox ⇒ From the **Capture mode** drop-down menu, select **all values**, as shown ⇒ Click **OK**.



Packet Delay Variation is the variance among end-to-end delays for voice packets received by this node.

Packet End-to-End Delay for a voice packet is measured from the time it is created to the time it is received.

5. Expand the **Voice Calling Party** hierarchy and select the following statistics: **Packet Delay Variation** and **Packet End-to-End Delay (sec)**.
6. Click **OK** and then save your project.

Voice_RSVP Called Statistics:

1. Right-click on the **Voice_RSVP Called** node and select **Choose Individual Statistics** from the pop-up menu.
2. Expand the **RSVP** hierarchy and select **Number of Resv States**.
3. Right-click on the **Number of Resv States** statistic ⇒ Select **Change Draw Style** from the pop-up menu ⇒ Choose **bar chart**.

4. Right-click on the **Number of Resv States** statistic ⇒ Select **Change Collection Mode** from the pop-up menu ⇒ Check the **Advanced** checkbox ⇒ From the **Capture mode** drop-down menu, select **all values** ⇒ Click **OK**.
5. Click **OK** and then save your project.

Voice Caller Statistics:

1. Right-click on the **Voice Caller** node and select **Choose Individual Statistics** from the pop-up menu.
2. Expand the **Voice Calling Party** hierarchy and select the following statistics: **Packet Delay Variation** and **Packet End-to-End Delay (sec)**
3. Click **OK** and then save your project.

Configure the Simulation

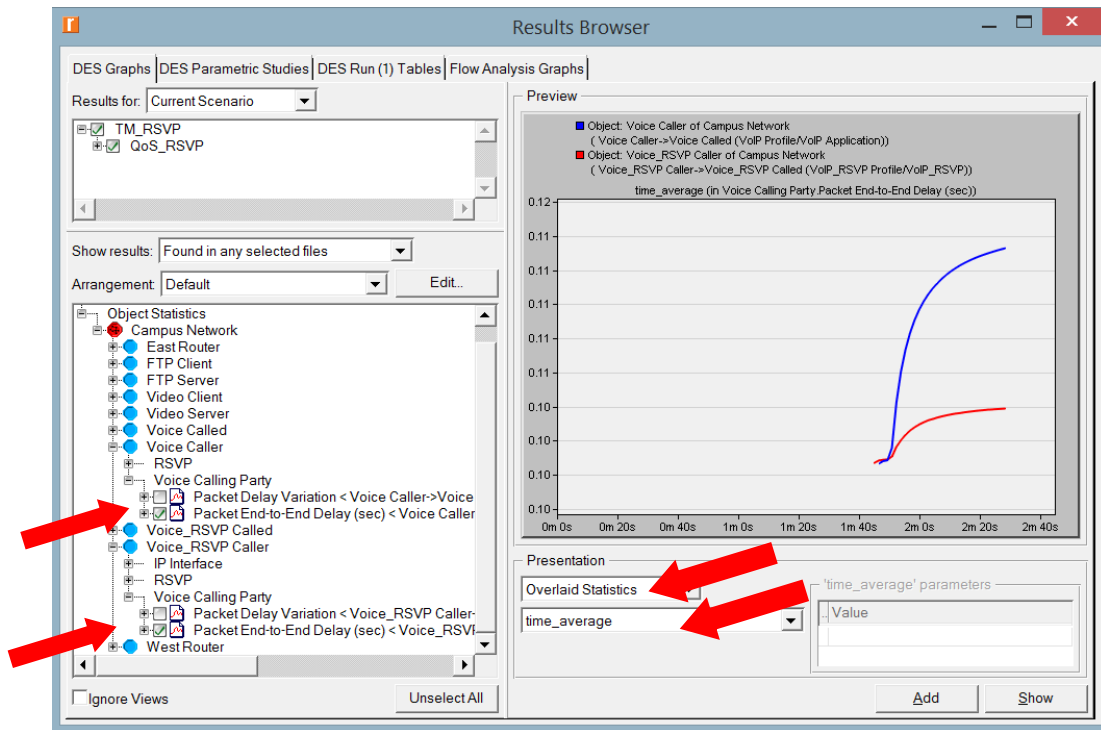
Here, we need to configure the duration of the simulation:

1. Click on  and the *Configure Simulation* window should appear.
2. Make sure that the duration is set to **150 seconds**.
3. Click **Run** and then save your project.

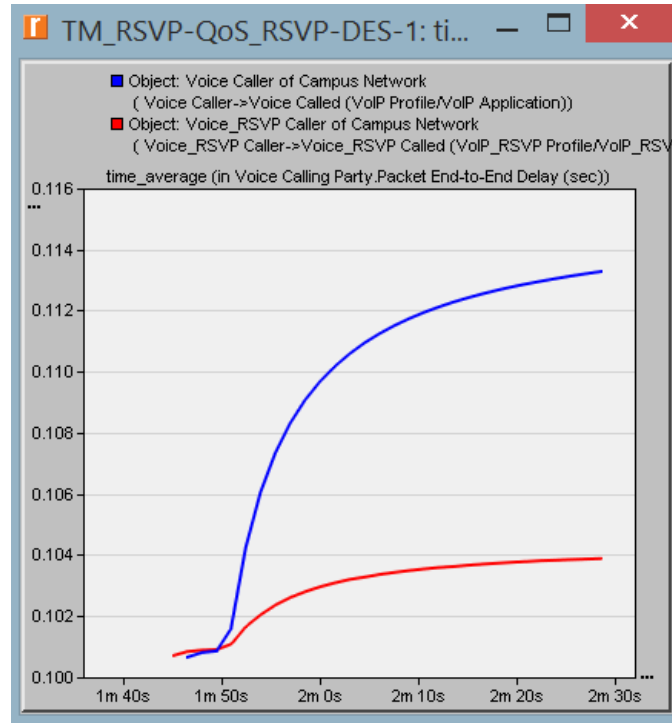
View the Results

To view and analyze the results:

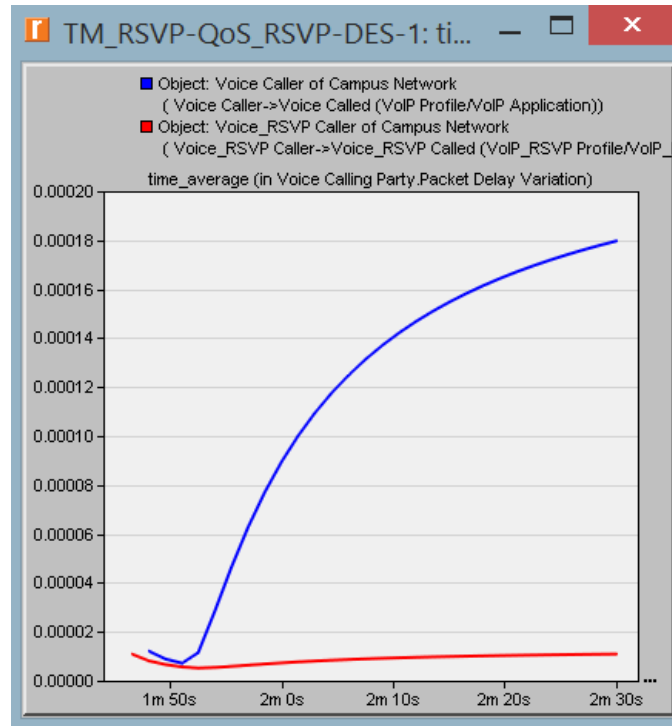
1. Select **View Results** from the **Results** menu.
2. As shown in the following figure, choose the **Packet End-to-End Delay** for both the **Voice Caller** and **Voice_RSVP Caller** nodes. Choose **Overlaid Statistics** and **time_average**.



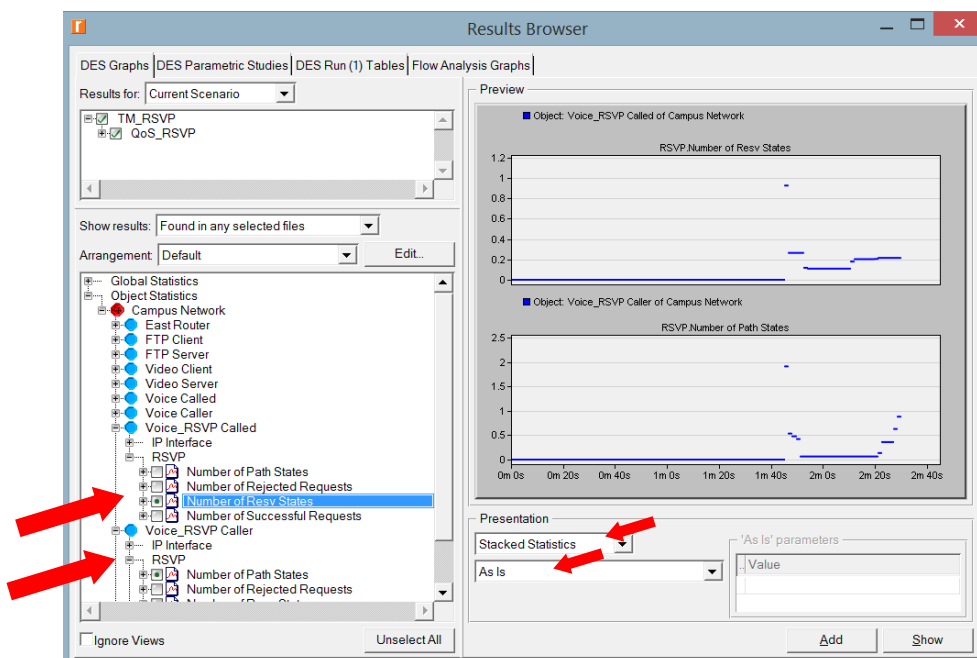
3. Click **Show** to get the following graph. (Note: To zoom in on the graph, click and drag your mouse to draw a rectangle around the area of interest and release the mouse button.)



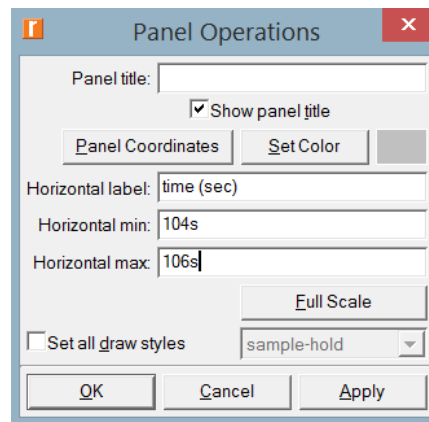
4. Similarly, you can get the following graph that compares the **Packet Delay Variation** for both the **Voice Caller** and **Voice_RSVP Caller** nodes. (Note: Make sure to “unselect” the statistics you chose for the previous graph.)



5. Finally, prepare the graph that displays the number of Path and Resv states by selecting the following statistics. Make sure to select **Stacked Statistics** and **As Is** as shown.

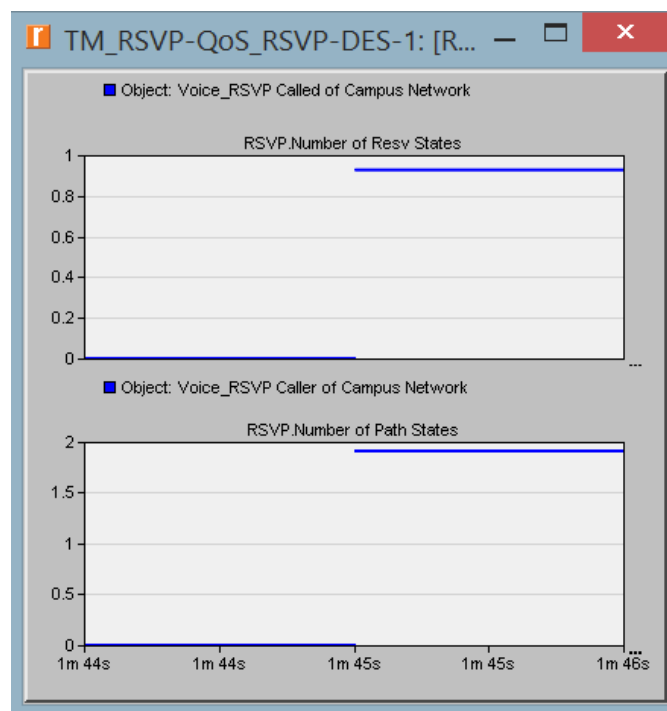


6. Right-click on the resulting graph and choose **Edit Panel Properties** ⇒ Change the assigned values to the **Horizontal Min** and **Horizontal Max** fields as shown (**your graph might require a different range**):



The 'Panel Operations' dialog box is shown. It has a title bar with a red 'X' button. Inside, there is a 'Panel title:' text box. Below it is a checked checkbox labeled 'Show panel title'. There are two buttons: 'Panel Coordinates' and 'Set Color'. Below these are three text boxes: 'Horizontal label:' with 'time (sec)', 'Horizontal min:' with '104s', and 'Horizontal max:' with '106s'. To the right of these is a 'Full Scale' button. Below the 'Horizontal min' and 'Horizontal max' boxes is a checkbox labeled 'Set all draw styles' and a dropdown menu showing 'sample-hold'. At the bottom are three buttons: 'OK', 'Cancel', and 'Apply'.

7. Click **OK**. The resulting graph should resemble the one below.



Questions

- 1) Analyze the graphs we obtained in this lab. Show the effect of RSVP on the Voice application and explain the obtained numbers of Path and Resv states.
- 2) How does the data rate of the link connecting the **East** and **West** routers affect the performance (e.g., **Packet End-to-End Delay**) of the Voice or Video Conference applications? To answer this question, create a new scenario as a duplicate of the **QoS_RSVP** scenario. Name the new scenario **Q2_HighRate**. In the **Q2_HighRate** scenario replace the current **PPP_DS1** link (data rate 1.544 Mbps) with a **PPP_DS3** link (data rate 44.736 Mbps).

Lab Report

Prepare a report that follows the guidelines explained in Lab 0. The report should include the answers to the above questions as well as the graphs you generated from the simulation scenarios. Discuss the results you obtained and compare these results with your expectations. Mention any anomalies or unexplained behaviors.